

The application of Topic Modelling in market segment identification of insurance companies



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Abstract

Natural language processing is one of the most popular subfields of Artificial Intelligence over the last few decades and will also be the most crucial development direction in the future. This technology has attracted more and more researchers to participate. Our industry partner, the Bank of England, also has a keen interest in this technology. They hope to integrate NLP technology into the work process and provide help for particular jobs.

In this paper, we discuss the application of topic modelling technology in extracting qualitative information and attempt to use the extracted information for further investigation on the insurance industry analyses.

Table of Contents

Abstract.....	2
1 Introduction	5
2 Data collection	6
3 Topic modelling	7
3.1 Algorithm.....	7
3.2 Modelling procedures	9
3.2.1 Analysing the sector-level data.....	9
3.2.2 Get the document-to-topic probabilities of news articles	13
4 Ensemble modelling (employment of topic modelling result).....	14
4.1 Enhance the data	14
4.2 Get document-to-topic probabilities of sub-sampling SFCR.....	15
4.3 Decision tree model training	15
4.3.1 Algorithm	16
4.3.2 Hyperparameters tuning.....	17
4.4 Predicting unlabelled SFCR	17
4.5 Model application	18
5 Conclusion	21
5.1 Limitation	21
5.2 Future application	21
References.....	22
Appendix.....	23

Figure 1 LDA assumption of a word-topic relationship in a document.....	7
Figure 2 Plate notation representing the LDA model	8
Figure 3 Example of Tokenise	10
Figure 4 Example of lemma.....	10
Figure 5 Topic Coherence curve	11
Figure 6 Inter-topic Distance Map via MDS of eight topics model.....	12
Figure 7 Trend of topics from 2014 to 2020	13
Figure 8 Flow chart of sub-sampling process	15
Figure 9 Quality comparison between boosting algorithm	16
Figure 10 Speed comparison between boosting algorithm	16
Figure 11 Models comparison.....	19
Figure 12 Model 1 predictions result summary	19
Figure 13 Model 2 predictions result summary.....	19
Figure 14 Model 3 predictions result summary	20

1 Introduction

The Bank of England is the UK's central bank. It has the mandate to promote the good of the people by maintaining monetary and financial stability. The Insurance Analytics Division (IAD) sits within the bank, which is responsible for the supervision of the Insurance sector. IAD supports supervision by developing tools that allow supervisors to understand sectoral risks and identify higher risk firms.

The IDA is now facing the problem of requiring a lot of text reading and spending much time to obtain qualitative information from reports. The qualitative information is used to evaluate the solvency risk for each firm. One of their jobs is to identify the market segment among the insurance industry for each firm as an essential procedure of the risk assessment.

Natural language processing is a technique that helps computers analyse human languages by processing the text into meaningful relationship and data, which is crucial for the interaction between the computer and human languages.

We have designed a solution using topic modelling technology for solving segment identification problems. Topic models are useful for analysing large collections of unlabelled text. It can quickly reflect the general content of the report by calculating the topic probabilities distribution in each report. Then use topic probabilities distribution as the features of the file, and achieve the purpose of identifying market segments through machine learning models such as decision tree.

2 Data collection

For the purpose of natural-language understanding on the firms reports, we needed to set up a corpus based on insurance industry information. Therefore, we selected ten insurance industry news websites and collected all the available news articles (over 13,000) from 2009 to 2020. These sector-level data are the training data for the topic clustering in the topic model.

We also collected the publicly released ‘Solvency and Financial Condition Report’ (SFCR) documents from the firms that are under the regulation of the Bank of England. SFCR is a comprehensive summary of the business performance, solvency risk, governance system and capital management of a company that are a crucial information source for the analyses of companies.

With the help of the Bank of England, we classified 115 SFCR documents from 35 companies into four market segments:

Class 1: London Market

Class 2: Retail

Class 3: Large Retail/Commercial

Class 4: Health

Sixteen SFCR documents from five companies were deliberately not classified: Allianz UK, AmTrust, Ecclesiastical, Flood Re and Allianz General. These 16 SFCR documents were taken as testing data, and are expected to be classified based on the model training of the 115 labelled SFCR documents.

3 Topic modelling

We chose to use the topic modelling method to analyse sector-level data, and we acquired a general understanding of the development direction and trend changes in this industry in the past ten years.

At the same time, the model established through sector-level data was applied to the SFCR reports of the insurance company, supervised by the Bank of England, to obtain the proportion of a given topic analysed by the model in a report.

3.1 Algorithm

We used the Latent Dirichlet Allocation (LDA) Mallet algorithm for topic modelling. MALLET is a Java-based package for statistical natural language processing, document classification, clustering, topic modelling, information mining and other machine learning applications to text. (McCallum, 2002)

The MALLET topic modelling toolkit contains efficient, sample-based implementations of Latent Dirichlet Allocation (LDA), one of the most common algorithms of topic modelling that can give the topic of each document in the form of a probability distribution. Moreover, it is an unsupervised learning algorithm. It does not need a manually labelled training set during training. All it requires is a document set and the number of specified latent topics. Besides, another advantage of LDA is that for each latent topic, some words can be found to describe it. (Blei D. , 2012)

LDA is a typical bag-of-words model: it considers a document to be a collection of a group of words, and there is no order or sequence relationship between words. A document can contain multiple topics, and each word in the document is generated from one of the topics. (H., P., H., & Vempala, 2000)

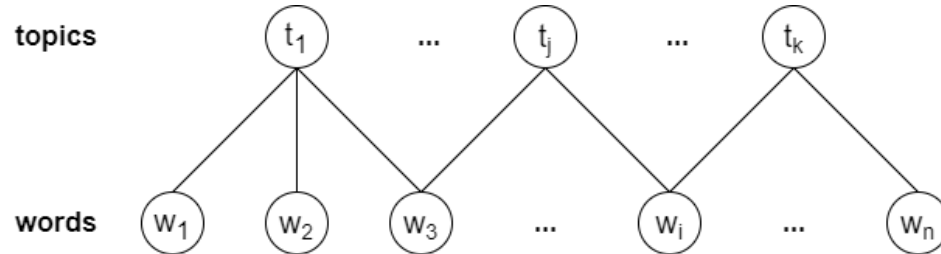


Figure 1 LDA assumption of a word-topic relationship in a document

The basic principle of the LDA model is to presume the document generative process in a Bayesian network structure and infer the latent topics in the corpus by reverse engineering. The LDA model has the following assumption of the document creation

(Blei, Ng, & Jordan, 2003):

For a corpus with V total words, established on M documents each of N_i words

- There are K topics across the corpus.
- Assume the topic probabilities θ_i of document i follows a Dirichlet distribution with parameter α , both θ_i and α are K -dimensional vectors.

$$\theta_i \sim \text{Dirichlet}_K(\alpha), \quad i \in \{1, \dots, M\}$$

- Assume the distribution of words in topic k is ϕ_k , which follows a Dirichlet distribution with parameter β , both ϕ_k and β are V -dimensional vectors.

$$\phi_k \sim \text{Dirichlet}_V(\beta), \quad k \in \{1, \dots, K\}$$

- The topic of the j^{th} word in document i is $z_{i,j}$, which follows one trial multinomial distribution (categorical distribution) with parameter θ_i , and $z_{i,j}$ could be integer from 1 to K .

$$z_{i,j} \sim \text{Categorical}_K(\theta_i)$$

- The j^{th} word in document i is $w_{i,j}$, which follows categorical distribution with parameter ϕ_k , and $w_{i,j}$ could be integer from 1 to V .

$$w_{i,j} \sim \text{Categorical}_V(\phi_k)$$

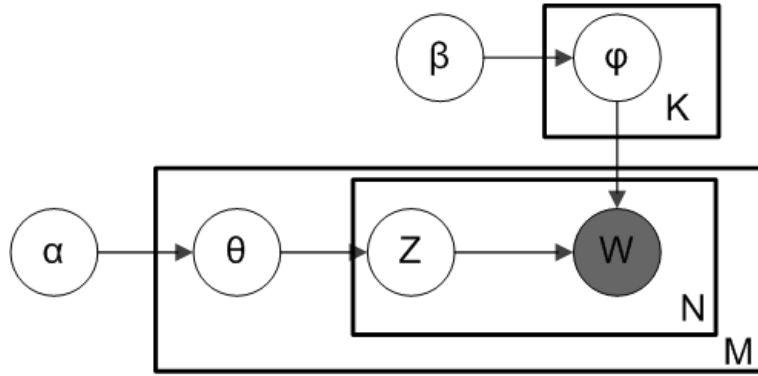


Figure 2 Plate notation representing the LDA model

Therefore, the joint distribution of all visible variables and hidden variables in the entire model is

$$p(w_i, z_i, \theta_i, \Phi | \alpha, \beta) = \prod_{j=1}^N p(\theta_i | \alpha) p(z_{i,j} | \theta_i) p(\Phi | \beta) p(w_{i,j} | \phi_{z_{i,j}})$$

By integrating θ_i and Φ of the above formula and summing z_i , we can get

$$p(w_i|\alpha, \beta) = \int_{\theta_i} \int_{\Phi} \sum_{z_i} p(w_i, z_i, \theta_i, \Phi|\alpha, \beta)$$

According to the maximum likelihood estimation of $p(w_i|\alpha, \beta)$, the parameters in the model can be estimated by Gibbs sampling and other methods. (Griffiths & Steyvers, 2004)

3.2 Modelling procedures

3.2.1 Analysing the sector-level data

i. Pre-processing of text

In order to perform the topic modelling, we needed to transform the complete articles into the corpus structure, which is adopted by the LDA model in computer languages like python. (Blei, Ng, & Jordan, 2003)

The pre-processing pipeline of the corpus transformation consists of the following parts:

- Define the stop-words

In human languages, there are some words that do not convey the main message in a sentence, that are also known as stop-words in Natural Languages Processing (NLP). These words can be safely removed without affecting the meaning of the sentence. The Gensim library provides common stop-words in English. Additionally, we defined a list of exclusive stop-words specific to the insurance field in order to obtain a clearer understanding of each clustered topic. We removed the names of websites, function words and common words to improve efficiency.

Common stop-words	Insurance stop-words
is	insurer
of	broker
and	insurance
...	...

Table 1 Example of stop-words

- Tokenise and lemma

The basic principle of topic modelling is to analyse the frequency of words. Therefore, we needed to cut the full text into a list of words, which is known as tokenisation.

Example for tokenization:

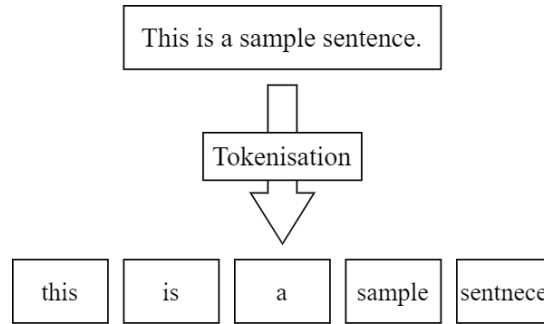
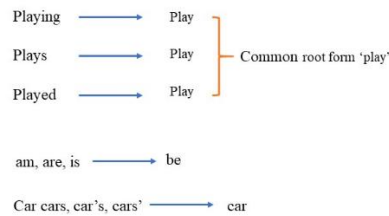


Figure 3 Example of Tokenise

Moreover, a word could have different forms, such as run, ran, running. These words have a similar meaning, but computers do not know their meaning and will treat them as entirely different words. It was essential to transform each word into its lemma, which is the base form of the word

Example for lemma:



Using above mapping a sentence could be normalized as follows:

the boy's cars are different colors → the boy car be differ color

Figure 4 Example of lemma

- Find phrase

Some words usually appear together, like ‘chief-executive’. This process will pair those kinds of words together. It will not only shorten the size of tokens but also help us understand the theme of topics better. ‘Chief-executive’ has a more explicit meaning than ‘chief’ or ‘executive’ that show up separately.

- ii. Determine the optimal number of retained topics

Determining the optimal number of topic clusters is a fundamental issue in topic modelling that requires the user to specify the number of topics to be generated so that the average topic coherence is maximised.

Topic Coherence evaluates a single topic by measuring the degree of semantic similarity between high scoring words in the topic. (Kapadia, 2019)

These measurements help distinguish between semantically interpretable topics and artefacts topics of statistical inference. In other words, a large topic coherence score typically means that the topic is more human- interpretable with less perplexity and should have a clearer view of the theme.

We used the concept of the elbow method for finding the optimal number of retained topics. The elbow method looks at the coherence value as a function of the number of topics: choose the number of topics so that adding another topic does not significantly improve the coherence score. (Thorndike, 1953)

Considering the lack of computing power and the time efficiency of the model evaluation, we evaluated the coherence score for each choice of topic numbered from one to 36 with an increment of five. As the chart below shows, we can see the average coherence scores are stabilised at 0.55 when the topic number is greater than five.

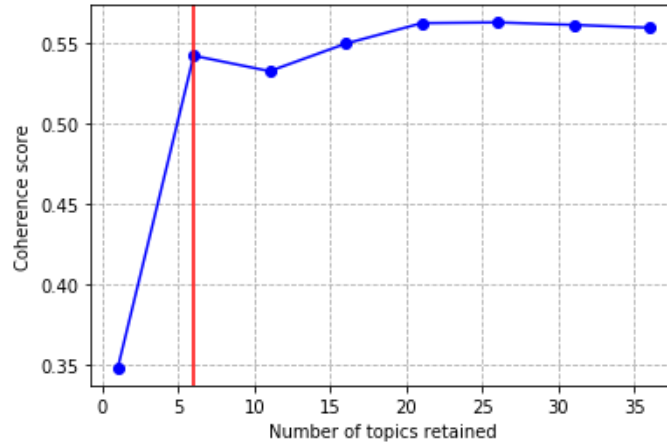


Figure 5 Topic Coherence curve

Therefore, we narrowed the range of retained topic numbers to five through eight. We plotted the Inter-topic Distance Map via multidimensional scaling (MDS) for each topic number, so that we can have a more intuitional observation of the similarity between topics.

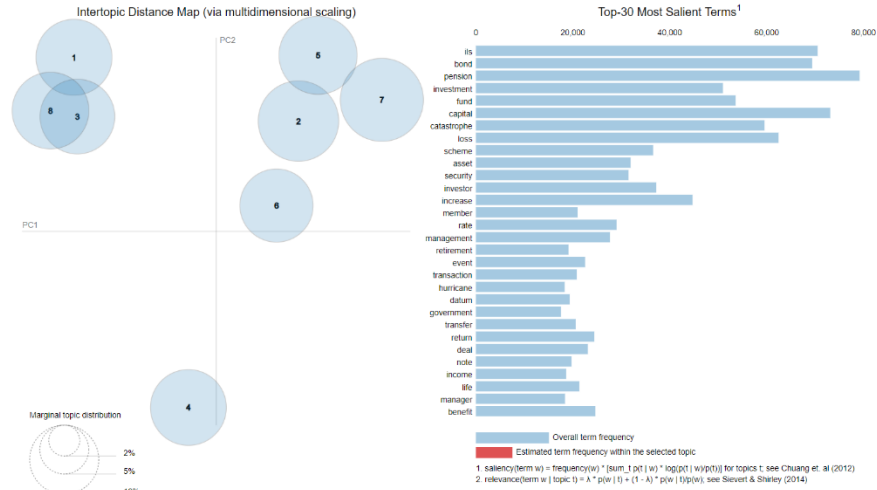


Figure 6 Inter-topic Distance Map via MDS of eight topics model

By comparing all the multidimensional scaling plots and referring to the recommendation from the Bank of England, the final decision of the retained topic number is eight.

The eight topics are:

- 1) pension, retirement, government, life,
- 2) capital, increase, rate
- 3) pension, scheme, member
- 4) client, role, team, solution
- 5) loss, event, hurricane
- 6) fund, investment, asset
- 7) ILS (insurance-linked securities), bond, catastrophe
- 8) datum, FCA (financial conduct authority), cyber

3.2.2 Get the document-to-topic probabilities of news articles

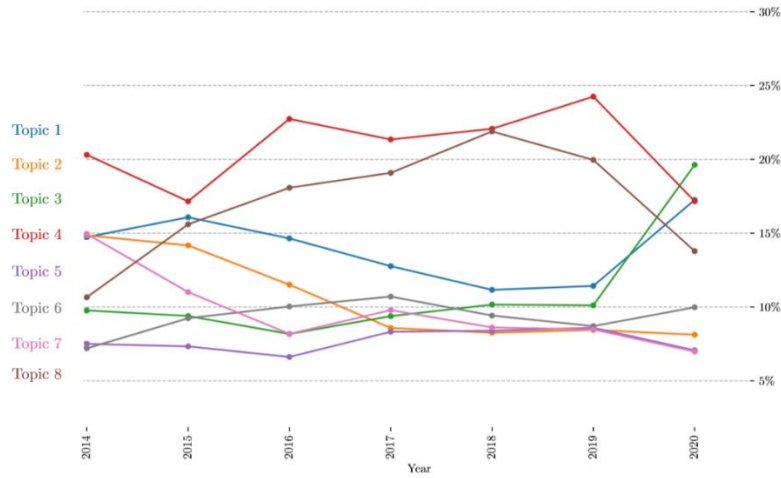


Figure 7 Trend of topics from 2014 to 2020

We employed the topic model on all new articles and calculated probabilities for topics by analysing the frequency of words. (McCallum, 2002) We then calculated the average topics probabilities distribution for all news articles in each year and plotted the line chart above. The line chart demonstrates the popularity trend of eight topics. We can see that topic 8, described by words datum and cyber, shows an upward trend from 2014 to 2018 and started to decline till now. Topic 1 and 3 have a very similar trend of popularity that is unsurprising as both topics are described by a similar set of words.

4 Ensemble modelling (employment of topic modelling result)

The ensemble modelling pipeline shown as following:

4.1 Enhance the data

The ulterior objective of the topic model is to generate valuable results from the SFCR in order to predict the market segment of firms using machine learning classification techniques. The number of labelled SFCR documents are insufficient for model training. A small training dataset will cause problems like overfitting and enlarging the impact of outliers.

To solve this issue, we increased our sample size by sub-sampling multiple times within each SFCR document. This enhanced the dataset by generating a larger sample size for training.

Sub-sampling procedures:

Step 1

We divided each SFCR into five sections according to the general structure used after the implication of Solvency II. The five sections are:

- A. BUSINESS AND PERFORMANCE
- B. SYSTEM OF GOVERNANCE
- C. RISK PROFILE
- D. VALUATION FOR SOLVENCY PURPOSES
- E. CAPITAL MANAGEMENT

Step 2

In each section, we performed the sampling technique on the basis of pages. The sampling size in each sampling is half of the total number of pages in each section. For example, if Section A of an SFCR document has X_1 pages, then we randomly choose $X_1/2$ pages without replacement from all pages in this section. We did the same thing to the rest of the sections and merged all five sections to form a sample SFCR document.

Step 3

By repeating Step 2 N -times with different random seeds, we generated N distinct sub-sampling examples from one original SFCR document. Sub-sampling examples are slightly different from each other because the randomly chosen pages in each repetition

of Step 2 are different.

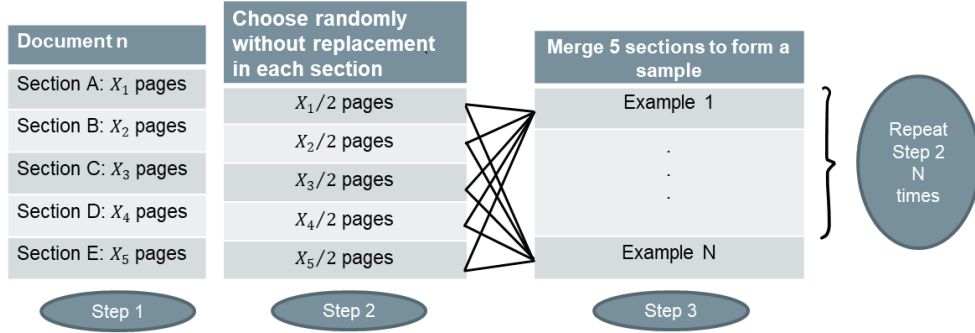


Figure 8 Flow chart of sub-sampling process

4.2 Get document-to-topic probabilities of sub-sampling SFCR

In the same way with the sector-level data, we needed to transform all the text in each SFCR sub-sampling example into a corpus. We employed the topic model and calculated probabilities for topics by analysing the frequency of words. By applying the pre-processing pipeline mentioned in 3.2.1 to the sub-sampling examples, we obtained the required corpus for calculating the topic probabilities for all examples.

probabilities	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8
$N \times$ SFCR sub-sampling examples	$p1_1$	$p2_1$	$p3_1$	$p4_1$	$p5_1$	$p6_1$	$p7_1$	$p8_1$
	...							
	...							
	$p1_N$	$p2_N$	$p3_N$	$p4_N$	$p5_N$	$p6_N$	$p7_N$	$p8_N$

Table 2 Example of the topic probabilities predictions

The topic probabilities demonstrate the general content structure of the SFCR examples in terms of the proportion of each topic taking place in the document.

More importantly, the eight topic probabilities could be taken as the features of the corresponding document and used for machine learning purposes.

We took the eight probabilities as the features of the corresponding document and applied the decision tree classification model.

4.3 Decision tree model training

In machine learning, decision tree is a predictive model that investigates the relationship

between the target value and the observational features of an item for classification or regression. Tree models where the target variable can take a discrete set of values are called classification trees. We used the labelled sub-sampling SFCR as the training data, topic probabilities as training features and market segments as target values.

4.3.1 Algorithm

CatBoost is one of the most powerful libraries for gradient boosting on decision tree. It is used by many giant companies and organizations like CERN, Cloudflare, Careem taxi, and in many fields such as weather prediction, recommendation systems, self-driving cars, or personal assistants. (Yandex, 2017)

CatBoost shows significant advantages in terms of the model prediction quality and learning speed compared to other boosting algorithms like LightGBM, XGBoost and H2O.

	CatBoost		LightGBM		XGBoost		H2O	
	Tuned	Default	Tuned	Default	Tuned	Default	Tuned	Default
 Adult	0.26974	0.27298 +1.21%	0.27602 +2.33%	0.28716 +6.46%	0.27542 +2.11%	0.28009 +3.84%	0.27510 +1.99%	0.27607 +2.35%
 Amazon	0.13772	0.13811 +0.29%	0.16360 +18.80%	0.16716 +21.38%	0.16327 +18.56%	0.16536 +20.07%	0.16264 +18.10%	0.16950 +23.08%
 KDD 98	0.19467	0.19479 +0.07%	0.19576 +0.56%	0.19837 +1.91%	0.19568 +0.52%	0.19795 +1.69%	0.19539 +0.37%	0.19607 +0.72%
 Kick prediction	0.28479	0.28491 +0.05%	0.29566 +3.82%	0.29877 +4.91%	0.29465 +3.47%	0.29816 +4.70%	0.29481 +3.52%	0.29635 +4.06%

Figures in this table represent Logloss values (lower is better) for Classification mode. Percentage is metric difference measured against tuned CatBoost results.

Figure 9 Quality comparison between boosting algorithm

	CatBoost	XGBoost	LightGBM
CPU (Xeon E5-2660v4)	527 sec	4339 sec	1146 sec
GTX 1080Ti (11GB)	18 sec	890 sec	110 sec

Dataset Epsilon (400K samples, 2000 features). Parameters: 128 bins, 64 leafs, 400 iterations.

Figure 10 Speed comparison between boosting algorithm

4.3.2 Hyperparameters tuning

In machine learning, a hyperparameter is a parameter whose value is used to control the learning process, which cannot be inferred with while fitting the machine to the training set. In contrast, the values of other parameters (typically node weights) are derived via training. A rational setting of hyperparameters can improve the speed and quality of the learning process. (J, T, M, DL, & R., 2019)

We used the grid search method for finding the optimal model parameters setting, that is achieved by the cooperation of CatBoost and the Hyperopt library in python. The built-in function in Hyperopt will automatically attempt all the possible combinations of hyper-parameters on a CatBoost cross-validation model within the parameter scope defined by us, and choose the best combination by minimising the loss function.

Hyperopt also allows us to define our own loss function. Instead of using validation accuracy only as the loss function, we also take the difference between training accuracy and validation accuracy into account for each CV model attempting.

$$\begin{aligned} overfit_1 &= \max(0, train_{acc} - val_{acc}) \\ overfit_2 &= \max(0, train_{acc} - val_{acc} - 0.05) \\ Loss\ function &= 1 - val_{acc} + overfit_1 \times 0.8 + overfit_2 \times 1 \end{aligned}$$

The term $overfit_1$ is the difference of training accuracy over validation accuracy. Additionally, overfitting greater than 5% is considered to be more dangerous, so we add the term $overfit_2$ which only includes the overfitting greater than 5%. By adding the two terms into the loss function, overfitting will be penalised when tuning the hyperparameters.

Calculate the accuracy of the optimal setting

After finding the optimal hyperparameters setting, we performed the CatBoost CV model again using the optimal setting, and calculated the average validation accuracy.

4.4 Predicting unlabelled SFCR

Similar to the labelled SFCR for model training, we sub-sampled N-times for unlabelled SFCR for predictions in order to enhance the dataset, and calculated the topic probabilities using the same procedure in 4.1 and 4.2.

It should be noted that a single decision tree cannot guarantee to return the globally optimal model due to the heuristic algorithms it is using, where locally optimal decisions

are made at each node. This can be mitigated by training multiple trees in an ensemble learner. (Ben-Gal, Dana, Shkolnik, & Singer, 2014)

We built five decision tree models using the same optimal hyperparameters settings and different random tree selections, and calculated the classification probabilities for each sample by averaging the result from each decision tree model. (Pedregosa, 2020)

4.5 Model application

Based on the ensemble model pipeline, we designed three different model applications for different purposes.

Model 1:

Use all labelled SFCR from 2016 to 2019 as the training data for the ensemble decision tree model, predict on unlabelled SFCR in 2018 and 2019 from the five investigating companies. Averaging prediction of all 2018 and 19 SFCR samples from a company (double weight for 2019 samples in averaging calculation), choose the segment with the highest probabilities as the final prediction.

Sub-sampling size N for both training and testing data = 10

Model 2:

Use labelled SFCR in 2018 and 2019 as the training data for the ensemble decision tree model, predict on unlabelled SFCR in 2018 and 2019 from the five investigating companies. Averaging prediction of all 2018 and 19 SFCR samples from a company (double weight for 2019 samples in averaging calculation), choose the segment with the highest probabilities as the final prediction.

Sub-sampling size N for both training and testing data = 20

Model 3:

Use labelled SFCR from 2016 to 2018 as the training data for the ensemble decision tree model, predict on SFCR in 2019. Averaging prediction of all 2019 SFCR samples from a company and choose the segment with the highest probabilities as the final prediction.

Sub-sampling size N for both training and testing data = 13

Both model 1 and model 2 are used for predicting unknown companies based on the

assumption that the market segment of a company is constant over time. Whereas, model 3 is designed for predicting the latest 2019 SFCR document, trying to investigate the change of market segment and emphasize the current situation of companies.

Results:

Model 1		Model 2		Model 3	
Training data	2016-19 labelled SFCR	Training data	2018-19 labelled SFCR	Training data	2016-18 labelled SFCR
Sub-sampling	10 times	Sub-sampling	20 times	Sub-sampling	13 times
Sample size	1090	Sample size	1060	Sample size	1118
Testing data	Unlabelled 2018-19 SFCR	Testing data	Unlabelled 2018-19 SFCR	Testing data	2019 SFCR
Training accuracy	81%	Training accuracy	85%	Training accuracy	87%

Figure 11 Models comparison

Looking at the cross-validation accuracy of the entire training data, all three models show a rational and satisfactory result around 85%, which means all three models are expected to generate valuable result without serious overfitting on the testing data.

It should be noted that we should also pay attention to the failed prediction in model3. The failed predictions might not necessarily be wrong, as we aim to predict the latest or future SFCR based on the past document record. The failed predictions might indicate the change of reporting style and the market segment of companies.

Model 1	Health	Large Retail/Commercial	London Market	Retail
Allianz UK	2.3%	26.2%	55.4%	16.2%
AmTrust Europe Ltd	2.8%	3.6%	64.8%	28.7%
Ecclesiastical Ins Office	2.7%	38.9%	38.6%	19.8%
FloodRe	3.5%	22.7%	26.8%	47.0%
Allianz Global	1.6%	17.8%	74.9%	5.7%

Figure 12 Model 1 predictions result summary

Model 2	Health	Large Retail/Commercial	London Market	Retail
Allianz UK	3.0%	21.9%	60.0%	15.2%
AmTrust Europe Ltd	3.6%	2.9%	74.9%	18.6%
Ecclesiastical Ins Office	3.1%	52.1%	30.6%	14.2%
FloodRe	5.2%	30.6%	23.0%	41.1%
Allianz Global	5.4%	16.7%	66.6%	11.4%

Figure 13 Model 2 predictions result summary

Comparing the prediction results on the unlabelled SFCR, we can see that model 1 and model 2 have the same market segment classification result. There is only a slight

difference in the segment probabilities. The impact of the variation of the training data in model 1 and 2 appears to be insignificant over our testing data.

Model 3	Health	Large Retail/Commercial	London Market	Retail	prediction	segment
Allianz Global	5.26%	10.52%	75.91%	8.31%	London Market	
ASPEN EU	1.13%	2.68%	91.62%	4.57%	London Market	London Market
AXA PPP	47.17%	42.99%	7.14%	2.70%	Health	Health
AXA UK	34.08%	56.86%	6.47%	2.58%	Large Retail/Commercial	Large Retail/Commercial
Ageas UK	5.11%	31.16%	6.05%	57.67%	Retail	Retail
Allianz UK	2.97%	19.70%	52.78%	24.55%	London Market	
AmTrust Europe Ltd	8.60%	9.93%	54.59%	26.88%	London Market	
Arch Ins Co UK	2.07%	6.75%	82.62%	8.56%	London Market	London Market
Aspen Ins UK	1.05%	2.41%	91.86%	4.68%	London Market	London Market
Aviva Insurance Ltd	6.55%	64.43%	6.78%	22.25%	Large Retail/Commercial	Large Retail/Commercial
BUPA Ins Ltd	50.64%	5.97%	25.31%	18.07%	Health	Health
DIRECT	6.20%	11.35%	27.49%	54.96%	Retail	London Market
Direct Line Group	4.71%	13.54%	20.81%	60.94%	Retail	Retail
Ecclesiastical Ins Office	4.03%	41.59%	27.02%	27.36%	Large Retail/Commercial	
Fidelis Underwriting Ltd	1.08%	1.62%	95.34%	1.95%	London Market	London Market
FloodRe	4.82%	51.29%	12.80%	31.10%	Large Retail/Commercial	
HISCOX	21.63%	11.30%	29.79%	37.28%	Retail	London Market
LV Group	33.91%	18.70%	26.24%	21.15%	Health	Retail
Lancashire UK	1.41%	4.05%	92.42%	2.12%	London Market	London Market
MMA	9.62%	23.38%	26.05%	40.94%	Retail	Retail
NORTH	5.75%	21.18%	56.31%	16.77%	London Market	London Market
QBE Europe	0.50%	0.92%	97.84%	0.75%	London Market	London Market
RSA Ins Group	2.17%	39.26%	51.88%	6.69%	London Market	Large Retail/Commercial
Vitality Health Ltd	81.71%	7.24%	4.36%	6.69%	Health	Health
WESTFIELD	60.93%	10.16%	16.46%	12.45%	Health	Health
XL	1.78%	5.72%	90.15%	2.35%	London Market	London Market
XL Catlin Ins Co UK Ltd	1.58%	3.44%	92.76%	2.22%	London Market	London Market
esure Group	8.32%	8.19%	22.65%	60.84%	Retail	Retail

Figure 14 Model 3 predictions result summary

The testing data prediction result from model 3 shows that four companies among the 23 labelled companies are classified to a different market segment compare to previous years. For investigating the change of company situation, these failed predictions could be the study focus.

The testing accuracy is 83% is only labelled companies are taking into account. To some extent, this result is matching the training accuracy 87%, which also means that overfitting is not significant in this model.

5 Conclusion

In conclusion, the application of topic modelling for classifying the market segment of insurance companies is feasible. By the use of the ensemble learning model based on the topic modelling result, reasonable prediction on the market segment of companies are generated, which could be a valuable reference for the analysis of an insurance company in the aspect of qualitative information.

5.1 Limitation

Due to the time of implementation of the Solvency ii policy, we can only find SFCR reports from 2016. Additionally, many insurance companies have not publicly published their SFCR as the regulation required, which leads to only very few SFCR documents could be found on the internet. As a result, we have to use the sub-sampling approach in order to have enough samples for model training. However, it will increase the covariance between documents as some are generating from the same source. The variety of training data remain at a lower level, which has impacted the quality of the trained model. The same problem also exists for testing data, where the sample size the testing data is not large enough for evaluating the performance of model 1 and 2 over the testing data.

5.2 Future application

Firstly, more SFCR document should be collected; a broader and diversified training data are expected to generate a better-quality model.

After having a larger enough testing data, model performance could be evaluated. Further investigation of the failed prediction could be done for the purpose of improving the performance of the model.

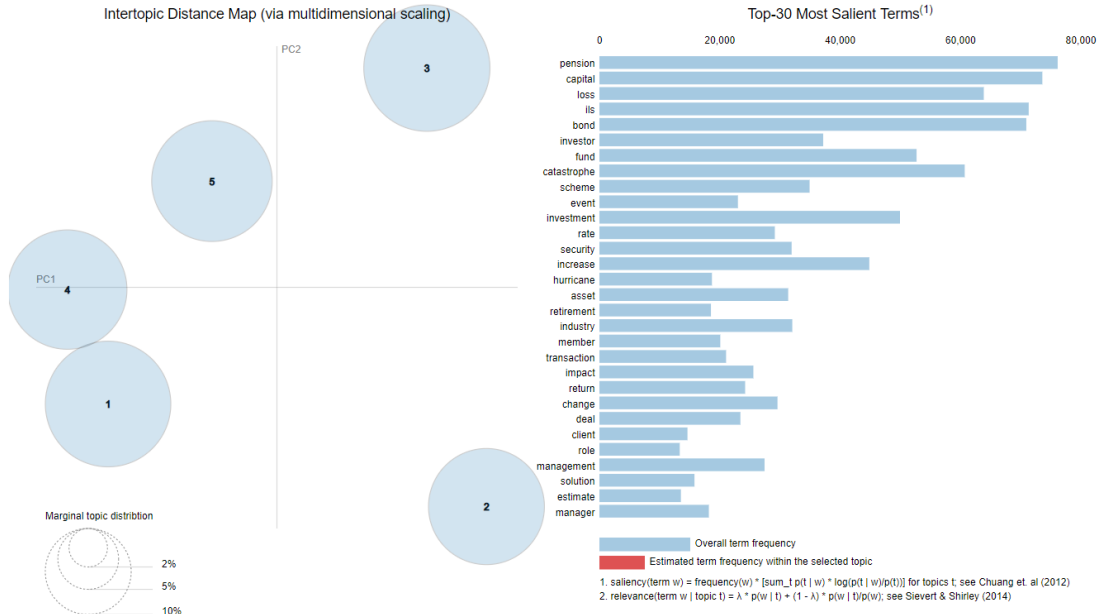
Moreover, the output of the topic modelling could be applied in other directions. For example, the relationship between the SFCR topics and the financial performance should also be considered. Research like this connect the qualitative information and quantitative information together, which might provide additional insight for analysing an insurance company.

References

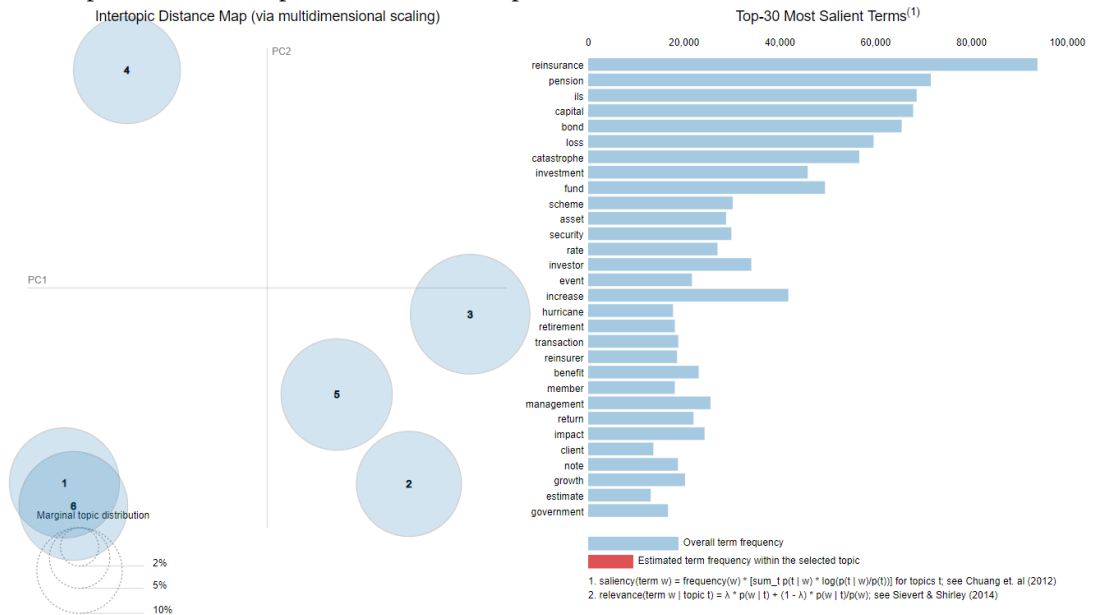
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Appendix

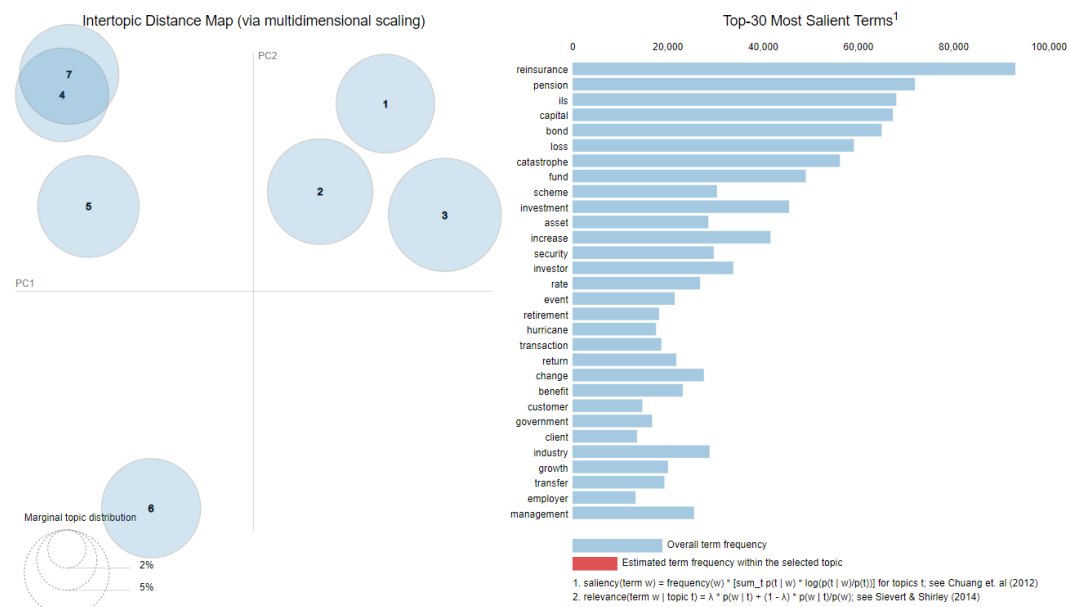
1. Inter-topic Distance Map via MDS of five topics model



2. Inter-topic Distance Map via MDS of six topics model



3. Inter-topic Distance Map via MDS of seven topics model



4. Model 1 prediction on testing data

company	year	sfc_r_id	Health	Large Retail/Commercial	London Market	Retail
Allianz UK	2019	10	3.42%	42.04%	30.21%	24.33%
Allianz UK	2019	10	2.98%	35.49%	53.46%	8.06%
Allianz UK	2019	10	0.93%	11.15%	83.03%	4.89%
Allianz UK	2019	10	1.37%	12.14%	80.86%	5.63%
Allianz UK	2019	10	2.46%	14.12%	41.33%	42.09%
Allianz UK	2019	10	2.33%	53.06%	30.94%	13.66%
Allianz UK	2019	10	1.54%	26.28%	65.89%	6.29%
Allianz UK	2019	10	1.31%	7.17%	59.94%	31.58%
Allianz UK	2019	10	3.82%	36.41%	44.77%	14.99%
Allianz UK	2019	10	2.62%	23.66%	63.48%	10.23%
AmTrust Europe Ltd	2018	13	0.29%	1.59%	95.38%	2.74%
AmTrust Europe Ltd	2018	13	0.51%	1.36%	90.05%	8.08%
AmTrust Europe Ltd	2018	13	3.81%	2.71%	53.60%	39.88%
AmTrust Europe Ltd	2018	13	0.82%	2.54%	60.78%	35.85%
AmTrust Europe Ltd	2018	13	1.17%	2.15%	85.35%	11.33%
AmTrust Europe Ltd	2018	13	0.62%	1.56%	92.77%	5.05%
AmTrust Europe Ltd	2018	13	1.39%	1.62%	68.74%	28.25%
AmTrust Europe Ltd	2018	13	2.25%	3.20%	70.17%	24.38%
AmTrust Europe Ltd	2018	13	0.16%	0.90%	98.16%	0.78%
AmTrust Europe Ltd	2018	13	0.15%	0.92%	98.13%	0.79%
AmTrust Europe Ltd	2019	14	4.45%	2.85%	40.62%	52.08%
AmTrust Europe Ltd	2019	14	2.69%	7.48%	36.95%	52.88%
AmTrust Europe Ltd	2019	14	2.43%	6.00%	55.97%	35.60%
AmTrust Europe Ltd	2019	14	1.48%	2.54%	65.02%	30.97%
AmTrust Europe Ltd	2019	14	1.72%	3.23%	75.38%	19.67%
AmTrust Europe Ltd	2019	14	3.28%	2.21%	65.98%	28.54%
AmTrust Europe Ltd	2019	14	6.00%	2.67%	47.13%	44.20%
AmTrust Europe Ltd	2019	14	1.08%	2.68%	54.77%	41.46%
AmTrust Europe Ltd	2019	14	3.96%	11.58%	58.02%	26.44%
AmTrust Europe Ltd	2019	14	10.04%	3.61%	65.70%	20.65%
Ecclesiastical Ins Office	2018	40	3.13%	43.75%	25.89%	27.23%
Ecclesiastical Ins Office	2018	40	2.60%	56.35%	18.89%	22.16%
Ecclesiastical Ins Office	2018	40	2.24%	37.99%	48.50%	11.27%
Ecclesiastical Ins Office	2018	40	1.97%	42.69%	45.44%	9.89%
Ecclesiastical Ins Office	2018	40	2.37%	26.60%	58.30%	12.73%
Ecclesiastical Ins Office	2018	40	2.48%	24.51%	60.59%	12.43%
Ecclesiastical Ins Office	2018	40	1.85%	20.69%	67.69%	9.77%
Ecclesiastical Ins Office	2018	40	2.93%	31.54%	52.46%	13.06%
Ecclesiastical Ins Office	2018	40	2.41%	54.38%	29.91%	13.29%
Ecclesiastical Ins Office	2018	40	2.66%	31.99%	42.11%	23.24%
Ecclesiastical Ins Office	2019	41	2.03%	21.42%	64.38%	12.16%
Ecclesiastical Ins Office	2019	41	3.02%	42.76%	26.11%	28.12%
Ecclesiastical Ins Office	2019	41	2.22%	39.35%	51.30%	7.13%
Ecclesiastical Ins Office	2019	41	2.49%	41.79%	22.82%	32.90%
Ecclesiastical Ins Office	2019	41	2.97%	25.37%	38.60%	33.06%
Ecclesiastical Ins Office	2019	41	2.41%	34.92%	40.17%	22.51%
Ecclesiastical Ins Office	2019	41	3.06%	39.76%	28.40%	28.78%
Ecclesiastical Ins Office	2019	41	2.25%	73.69%	15.24%	8.82%
Ecclesiastical Ins Office	2019	41	4.04%	44.76%	20.20%	31.00%
Ecclesiastical Ins Office	2019	41	2.97%	34.25%	47.45%	15.33%
FloodRe	2018	55	2.65%	8.33%	19.68%	69.35%
FloodRe	2018	55	4.32%	55.46%	17.49%	22.73%
FloodRe	2018	55	3.41%	41.60%	25.23%	29.76%
FloodRe	2018	55	2.28%	11.62%	16.76%	69.34%
FloodRe	2018	55	2.00%	7.68%	11.25%	79.07%
FloodRe	2018	55	4.02%	40.69%	38.22%	17.07%
FloodRe	2018	55	5.53%	26.37%	31.22%	36.89%
FloodRe	2018	55	2.37%	13.83%	8.92%	74.88%
FloodRe	2018	55	3.02%	19.87%	67.02%	10.09%
FloodRe	2018	55	5.01%	26.39%	22.19%	46.41%
FloodRe	2019	56	4.97%	46.40%	6.97%	41.65%
FloodRe	2019	56	3.10%	27.84%	5.86%	63.20%
FloodRe	2019	56	3.48%	27.78%	35.84%	32.90%
FloodRe	2019	56	3.50%	13.37%	7.03%	76.10%
FloodRe	2019	56	3.17%	9.02%	36.58%	51.23%
FloodRe	2019	56	2.04%	7.64%	45.02%	45.29%
FloodRe	2019	56	6.87%	33.10%	33.25%	26.79%
FloodRe	2019	56	2.59%	14.30%	12.76%	70.36%
FloodRe	2019	56	3.47%	24.58%	41.21%	30.75%
FloodRe	2019	56	2.23%	10.55%	48.57%	38.65%
AGCS	2019	145	1.73%	20.20%	71.29%	6.78%
AGCS	2019	145	1.82%	19.11%	73.73%	5.34%
AGCS	2019	145	1.24%	17.47%	76.73%	4.56%
AGCS	2019	145	1.74%	19.53%	71.77%	6.97%
AGCS	2019	145	0.71%	10.18%	86.53%	2.57%
AGCS	2019	145	1.27%	16.71%	77.13%	4.89%
AGCS	2019	145	1.87%	18.69%	73.22%	6.22%
AGCS	2019	145	1.72%	16.05%	77.07%	5.16%
AGCS	2019	145	1.70%	19.71%	71.52%	7.07%
AGCS	2019	145	1.81%	20.29%	70.46%	7.44%

5. Model 2 prediction on testing data

company	year	ifcc id	Health	Large Retail/Commercial	London Market	Rest
Alliana UK	2019	10	2.30%	18.92%	59.40%	19.39%
Alliana UK	2019	10	3.13%	11.72%	67.47%	17.68%
Alliana UK	2019	10	1.60%	12.93%	76.82%	8.60%
Alliana UK	2019	10	5.72%	16.23%	52.11%	25.93%
Alliana UK	2019	10	0.99%	5.14%	89.24%	5.07%
Alliana UK	2019	10	2.22%	13.37%	78.51%	5.91%
Alliana UK	2019	10	4.49%	19.05%	53.54%	22.93%
Alliana UK	2019	10	3.37%	46.47%	27.15%	23.01%
Alliana UK	2019	10	0.62%	5.66%	91.49%	2.23%
Alliana UK	2019	10	6.62%	8.63%	44.77%	39.98%
Alliana UK	2019	10	1.77%	37.98%	54.37%	6.38%
Alliana UK	2019	10	2.42%	14.62%	73.23%	9.74%
Alliana UK	2019	10	4.56%	18.79%	56.05%	20.59%
Alliana UK	2019	10	2.12%	66.81%	20.55%	10.52%
Alliana UK	2019	10	2.24%	13.27%	73.61%	10.88%
Alliana UK	2019	10	0.99%	3.24%	71.33%	24.45%
Alliana UK	2019	10	5.88%	77.31%	56.30%	10.13%
Alliana UK	2019	10	3.00%	15.59%	67.97%	13.44%
Alliana UK	2019	10	1.28%	63.99%	28.48%	6.25%
Alliana UK	2019	10	5.17%	16.30%	56.03%	21.51%
AmTrust Europe Ltd	2018	13	2.48%	2.78%	84.20%	10.54%
AmTrust Europe Ltd	2018	13	2.14%	2.83%	81.78%	13.29%
AmTrust Europe Ltd	2018	13	1.53%	2.54%	82.06%	13.87%
AmTrust Europe Ltd	2018	13	0.96%	1.75%	89.15%	8.14%
AmTrust Europe Ltd	2018	13	11.49%	5.95%	51.62%	30.55%
AmTrust Europe Ltd	2018	13	1.81%	2.77%	77.50%	17.92%
AmTrust Europe Ltd	2018	13	1.17%	2.64%	84.50%	11.68%
AmTrust Europe Ltd	2018	13	0.28%	1.61%	91.62%	5.80%
AmTrust Europe Ltd	2018	13	1.38%	1.91%	86.04%	10.67%
AmTrust Europe Ltd	2018	13	2.17%	3.34%	89.96%	4.53%
AmTrust Europe Ltd	2018	13	0.25%	0.70%	97.78%	1.26%
AmTrust Europe Ltd	2018	13	1.01%	1.61%	85.04%	12.33%
AmTrust Europe Ltd	2018	13	1.80%	2.10%	81.52%	14.58%
AmTrust Europe Ltd	2018	13	1.09%	1.37%	92.99%	4.40%
AmTrust Europe Ltd	2018	13	2.23%	2.76%	76.50%	18.50%
AmTrust Europe Ltd	2018	13	0.83%	1.46%	91.82%	5.90%
AmTrust Europe Ltd	2018	13	1.22%	3.22%	88.13%	7.23%
AmTrust Europe Ltd	2018	13	2.73%	2.47%	65.31%	29.49%
AmTrust Europe Ltd	2018	13	1.72%	2.47%	86.08%	9.73%
AmTrust Europe Ltd	2018	13	0.41%	0.79%	96.52%	2.08%
AmTrust Europe Ltd	2019	14	3.13%	1.93%	65.64%	29.30%
AmTrust Europe Ltd	2019	14	3.74%	3.14%	71.86%	21.26%
AmTrust Europe Ltd	2019	14	22.11%	4.48%	33.53%	58.89%
AmTrust Europe Ltd	2019	14	3.62%	3.63%	55.36%	37.39%
AmTrust Europe Ltd	2019	14	1.00%	1.62%	81.85%	15.47%
AmTrust Europe Ltd	2019	14	2.09%	8.92%	64.26%	23.83%
AmTrust Europe Ltd	2019	14	3.23%	2.30%	44.45%	50.02%
AmTrust Europe Ltd	2019	14	10.93%	5.18%	24.14%	59.76%
AmTrust Europe Ltd	2019	14	4.08%	2.38%	54.19%	38.96%
AmTrust Europe Ltd	2019	14	3.97%	3.20%	66.14%	26.69%
AmTrust Europe Ltd	2019	14	4.98%	2.12%	34.37%	58.53%
AmTrust Europe Ltd	2019	14	8.61%	6.60%	64.37%	20.37%
AmTrust Europe Ltd	2019	14	4.40%	3.59%	41.47%	50.34%
AmTrust Europe Ltd	2019	14	1.17%	1.99%	68.61%	27.03%
AmTrust Europe Ltd	2019	14	5.05%	5.99%	56.75%	32.21%
AmTrust Europe Ltd	2019	14	21.08%	6.91%	42.08%	29.92%
AmTrust Europe Ltd	2019	14	14.25%	7.09%	46.52%	22.38%
AmTrust Europe Ltd	2019	14	4.21%	4.55%	63.02%	28.21%
AmTrust Europe Ltd	2019	14	2.00%	3.12%	66.66%	28.21%
AmTrust Europe Ltd	2019	14	0.98%	1.42%	67.62%	9.77%
Eccelesiastical Ins Office	2018	40	3.76%	76.63%	11.46%	8.14%
Eccelesiastical Ins Office	2018	40	1.73%	78.43%	12.59%	7.24%
Eccelesiastical Ins Office	2018	40	3.54%	26.44%	57.05%	12.95%
Eccelesiastical Ins Office	2018	40	1.94%	47.06%	43.04%	7.96%
Eccelesiastical Ins Office	2018	40	0.62%	5.80%	86.74%	6.64%
Eccelesiastical Ins Office	2018	40	1.48%	52.17%	67.83%	11.11%
Eccelesiastical Ins Office	2018	40	1.47%	45.51%	45.94%	7.08%
Eccelesiastical Ins Office	2018	40	1.96%	48.94%	41.90%	7.20%
Eccelesiastical Ins Office	2018	40	2.78%	70.45%	14.76%	10.01%
Eccelesiastical Ins Office	2018	40	2.48%	73.83%	12.96%	10.72%
Eccelesiastical Ins Office	2018	40	3.68%	63.70%	18.21%	14.41%
Eccelesiastical Ins Office	2018	40	2.81%	28.26%	20.50%	48.43%
Eccelesiastical Ins Office	2018	40	1.88%	75.61%	13.00%	9.53%
Eccelesiastical Ins Office	2018	40	4.07%	21.74%	59.04%	15.19%
Eccelesiastical Ins Office	2018	40	7.03%	65.38%	17.38%	10.01%
Eccelesiastical Ins Office	2018	40	2.10%	32.10%	54.01%	11.79%
Eccelesiastical Ins Office	2018	40	3.84%	58.65%	12.25%	15.26%
Eccelesiastical Ins Office	2018	40	3.33%	51.88%	51.88%	15.50%
Eccelesiastical Ins Office	2018	40	1.54%	52.72%	37.62%	8.12%
Eccelesiastical Ins Office	2018	40	2.97%	56.16%	30.24%	20.64%
Eccelesiastical Ins Office	2019	41	2.17%	64.24%	21.39%	12.19%
Eccelesiastical Ins Office	2019	41	4.92%	27.10%	33.68%	34.30%
Eccelesiastical Ins Office	2019	41	3.52%	81.20%	6.62%	4.68%
Eccelesiastical Ins Office	2019	41	3.26%	56.40%	11.97%	28.32%
Eccelesiastical Ins Office	2019	41	4.13%	69.28%	13.48%	13.11%
Eccelesiastical Ins Office	2019	41	3.13%	27.52%	51.12%	14.02%
Eccelesiastical Ins Office	2019	41	3.18%	66.66%	20.29%	9.89%
Eccelesiastical Ins Office	2019	41	2.20%	69.33%	18.29%	10.18%
Eccelesiastical Ins Office	2019	41	2.40%	57.44%	30.87%	9.29%
Eccelesiastical Ins Office	2019	41	3.56%	61.52%	21.63%	13.29%
Eccelesiastical Ins Office	2019	41	4.08%	28.92%	40.58%	17.42%
Eccelesiastical Ins Office	2019	41	2.45%	79.15%	9.47%	8.93%
Eccelesiastical Ins Office	2019	41	2.42%	60.69%	22.28%	14.62%
Eccelesiastical Ins Office	2019	41	6.80%	47.34%	23.74%	22.12%
Eccelesiastical Ins Office	2019	41	2.90%	69.06%	17.29%	11.59%
Eccelesiastical Ins Office	2019	41	12.30%	29.72%	40.19%	17.79%
Eccelesiastical Ins Office	2019	41	4.45%	25.06%	24.73%	45.76%
Eccelesiastical Ins Office	2019	41	3.43%	58.44%	8.33%	29.81%
Eccelesiastical Ins Office	2019	41	2.97%	38.72%	31.54%	26.77%
Eccelesiastical Ins Office	2019	41	1.91%	63.72%	10.31%	14.99%
FloodRe	2018	55	3.26%	17.40%	60.75%	19.08%
FloodRe	2018	55	3.54%	35.34%	16.12%	45.00%
FloodRe	2018	55	8.40%	12.29%	19.49%	59.76%
FloodRe	2018	55	1.93%	2.18%	9.16%	88.76%
FloodRe	2018	55	6.39%	20.80%	16.00%	56.81%
FloodRe	2018	55	4.81%	30.58%	21.59%	43.03%
FloodRe	2018	55	5.47%	56.98%	24.08%	11.47%
FloodRe	2018	55	4.80%	50.28%	25.72%	19.20%
FloodRe	2018	55	5.13%	10.15%	10.42%	74.29%
FloodRe	2018	55	5.31%	55.44%	20.51%	19.74%
FloodRe	2018	55	4.98%	16.89%	31.68%	46.44%
FloodRe	2018	55	7.73%	34.82%	12.31%	45.14%
FloodRe	2018	55	7.27%	44.50%	14.76%	33.81%
FloodRe	2018	55	5.08%	19.51%	22.35%	53.06%
FloodRe	2018	55	5.05%	29.04%	27.10%	28.81%
FloodRe	2018	55	5.22%	24.94%	6.16%	63.67%
FloodRe	2018	55	8.75%	24.65%	12.44%	54.16%
FloodRe	2018	55	4.29%	19.67%	54.68%	21.36%
FloodRe	2018	55	2.09%	2.33%	12.83%	82.76%
FloodRe	2018	55	4.08%	6.63%	14.99%	74.30%
FloodRe	2018	56	2.88%	55.26%	21.04%	1.79%
FloodRe	2018	56	5.02%	27.72%	40.27%	26.99%
FloodRe	2018	56	7.26%	41.87%	27.22%	23.65%
FloodRe	2018	56	8.00%	28.88%	30.80%	32.25%
FloodRe	2018	56	2.66%	82.48%	6.17%	8.69%
FloodRe	2018	56	7.02%	40.51%	26.94%	25.53%
FloodRe	2018	56	6.10%	46.04%	26.18%	19.62%
FloodRe	2018	56	5.60%	40.33%	15.19%	38.88%
FloodRe	2018	56	6.50%	46.80%	10.05%	26.65%
FloodRe	2018	56	6.23%	27.40%	30.46%	35.91%
FloodRe	2018	56	3.53%	9.97%	23.63%	62.88%
FloodRe	2018	56	4.40%	63.08%	12.82%	19.49%
FloodRe	2018	56	2.78%	6.19%	59.10%	31.92%
FloodRe	2018	56	5.19%	50.03%	20.85%	23.93%
FloodRe	2018	56	5.23%	66.87%	11.50%	16.40%
FloodRe	2018	56	4.56%	20.06%	34.70%	40.68%
FloodRe	2018	56	6.56%	34.55%	22.40%	36.49%
FloodRe	2018	56	6.38%	45.15%	24.78%	21.69%
FloodRe	2018	56	4.78%	54.26%	23.35%	17.60%
FloodRe	2018	56	5.08%	19.45%	23.54%	45.34%
AGCS	2019	145	7.29%	14.10%	68.90%	9.71%
AGCS	2019	145	5.31%	20.85%	61.54%	12.30%
AGCS	2019	145	4.33%	10.99%	76.34%	8.34%
AGCS	2019	145	7.80%	19.76%	59.80%	12.61%
AGCS	2019	145	6.36%	43.58%	30.22%	19.85%
AGCS	2019	145	6.23%	12.90%	71.66%	9.21%
AGCS	2019	145	4.76%	20.61%	65.02%	11.61%
AGCS	2019	145	6.90%	20.06%	58.99%	14.06%
AGCS	2019	145	5.05%	15.72%	64.84%	14.40%
AGCS	2019	145	3.97%	9.35%	79.31%	9.37%
AGCS	2019	145	5.71%	16.29%	64.97%	13.08%
AGCS	2019	145	6.22%	17.84%	61.00%	14.94%
AGCS	2019	145	4.17%	9.03%	78.37%	8.23%
AGCS	2019	145	4.00%	17.15%	68.41%	10.44%
AGCS	2019	145	6.13%	16.44%	66.26%	11.98%
AGCS	2019	145	6.47%	19.23%	62.12%	17.98%
AGCS	2019	145	4.88%	16.04%	71.19%	7.89%
AGCS	2019	145	3.83%	9.07%	79.81%	7.29%
AGCS	2019	145	5.51%	17.44%	63.89%	12.96%
AGCS	2019	145	2.43%	6.54%	84.80%	6.22%

6. Model 3 prediction on testing data

company_id	year	stcr_id	Health	Large Retail/Commercial	London Market	Retail
Agasus UK	2019	3	2.801%	74.91%	3.11%	19.17%
Agasus UK	2019	3	10.90%	17.44%	15.61%	16.39%
Agasus UK	2019	3	6.22%	20.81%	12.02%	60.90%
Agasus UK	2019	3	3.10%	22.72%	4.63%	69.29%
Agasus UK	2019	3	5.96%	17.68%	3.62%	72.76%
Agasus UK	2019	3	5.29%	16.52%	5.70%	72.44%
Agasus UK	2019	3	5.25%	15.39%	4.26%	75.39%
Agasus UK	2019	3	3.28%	59.96%	3.90%	33.40%
Agasus UK	2019	3	3.10%	74.21%	2.05%	20.64%
Agasus UK	2019	3	3.44%	86.48%	4.15%	75.95%
Agasus UK	2019	3	6.45%	19.48%	11.90%	62.19%
Agasus UK	2019	3	6.27%	37.12%	5.24%	52.66%
Agasus UK	2019	3	4.65%	12.35%	4.92%	78.69%
Allianz UK	2019	10	0.69%	1.22%	96.92%	1.17%
Allianz UK	2019	10	2.98%	7.79%	82.06%	7.58%
Allianz UK	2019	10	3.08%	46.13%	38.22%	12.39%
Allianz UK	2019	10	2.47%	36.96%	22.02%	38.56%
Allianz UK	2019	10	2.25%	3.89%	85.78%	8.18%
Allianz UK	2019	10	3.41%	13.39%	75.08%	8.12%
Allianz UK	2019	10	4.35%	39.09%	25.12%	31.55%
Allianz UK	2019	10	2.81%	27.40%	51.56%	18.24%
Allianz UK	2019	10	2.40%	10.11%	77.19%	10.39%
Allianz UK	2019	10	4.98%	8.89%	5.12%	81.19%
Allianz UK	2019	10	3.39%	36.83%	24.66%	35.12%
Allianz UK	2019	10	4.14%	21.73%	10.61%	63.47%
Allianz UK	2019	10	1.53%	2.71%	92.56%	9.40%
AmTrust Europe Ltd	2019	14	8.22%	7.05%	18.68%	66.39%
AmTrust Europe Ltd	2019	14	9.88%	9.79%	52.64%	27.69%
AmTrust Europe Ltd	2019	14	7.62%	10.77%	66.62%	14.99%
AmTrust Europe Ltd	2019	14	9.52%	11.25%	63.89%	15.75%
AmTrust Europe Ltd	2019	14	5.08%	8.61%	44.39%	41.94%
AmTrust Europe Ltd	2019	14	8.48%	12.28%	61.59%	17.64%
AmTrust Europe Ltd	2019	14	1.67%	4.57%	94.73%	8.99%
AmTrust Europe Ltd	2019	14	6.58%	9.22%	40.27%	43.95%
AmTrust Europe Ltd	2019	14	6.38%	10.42%	55.26%	27.99%
AmTrust Europe Ltd	2019	14	12.70%	15.11%	39.82%	31.97%
AmTrust Europe Ltd	2019	14	19.05%	9.89%	52.38%	18.68%
AmTrust Europe Ltd	2019	14	2.69%	4.88%	84.14%	8.99%
AmTrust Europe Ltd	2019	14	15.05%	14.84%	45.26%	24.85%
Arch Ins Co UK	2019	17	0.98%	3.94%	90.99%	4.09%
Arch Ins Co UK	2019	17	2.89%	7.89%	85.50%	4.21%
Arch Ins Co UK	2019	17	2.37%	24.25%	63.23%	10.15%
Arch Ins Co UK	2019	17	0.97%	1.36%	94.97%	2.71%
Arch Ins Co UK	2019	17	0.47%	1.21%	97.08%	1.21%
Arch Ins Co UK	2019	17	0.92%	1.99%	92.60%	4.49%
Arch Ins Co UK	2019	17	0.26%	0.98%	97.69%	1.02%
Arch Ins Co UK	2019	17	0.24%	0.48%	98.78%	0.51%
Arch Ins Co UK	2019	17	2.65%	14.59%	72.11%	10.59%
Arch Ins Co UK	2019	17	4.92%	11.36%	67.61%	16.12%
Arch Ins Co UK	2019	17	3.00%	3.99%	86.54%	5.03%
Arch Ins Co UK	2019	17	0.89%	1.46%	96.61%	1.84%
Arch Ins Co UK	2019	17	7.12%	14.38%	31.00%	47.51%
Agasus Ins UK	2019	21	0.46%	0.76%	97.91%	0.87%
Agasus Ins UK	2019	21	1.15%	5.14%	88.67%	5.24%
Agasus Ins UK	2019	21	0.47%	0.73%	98.02%	0.79%
Agasus Ins UK	2019	21	3.82%	10.13%	60.44%	25.60%
Agasus Ins UK	2019	21	0.41%	0.81%	98.22%	0.77%
Agasus Ins UK	2019	21	0.24%	0.39%	99.12%	0.34%
Agasus Ins UK	2019	21	0.45%	1.02%	97.13%	1.00%
Agasus Ins UK	2019	21	1.00%	1.61%	94.76%	2.63%
Agasus Ins UK	2019	21	0.25%	0.45%	98.75%	0.55%
Agasus Ins UK	2019	21	0.62%	1.09%	96.75%	1.64%
Agasus Ins UK	2019	21	1.12%	2.72%	91.25%	4.91%
Agasus Ins UK	2019	21	0.48%	0.96%	97.20%	1.42%
Agasus Ins UK	2019	21	1.12%	5.61%	75.62%	15.22%
Aviva Insurance Ltd	2019	25	2.43%	80.82%	3.94%	12.83%
Aviva Insurance Ltd	2019	25	3.55%	39.26%	7.12%	40.79%
Aviva Insurance Ltd	2019	25	2.52%	76.07%	3.63%	17.78%
Aviva Insurance Ltd	2019	25	15.44%	54.18%	9.53%	20.83%
Aviva Insurance Ltd	2019	25	2.10%	86.33%	1.73%	8.18%
Aviva Insurance Ltd	2019	25	1.78%	86.89%	1.49%	9.84%
Aviva Insurance Ltd	2019	25	19.23%	12.71%	13.11%	14.49%
Aviva Insurance Ltd	2019	25	12.39%	44.48%	7.26%	35.87%
Aviva Insurance Ltd	2019	25	4.16%	78.62%	2.72%	14.51%
Aviva Insurance Ltd	2019	25	2.58%	85.38%	4.47%	7.75%
Aviva Insurance Ltd	2019	25	4.19%	19.74%	8.68%	67.39%
Aviva Insurance Ltd	2019	25	11.84%	55.70%	21.64%	10.62%
Aviva Insurance Ltd	2019	25	1.51%	77.20%	2.51%	18.54%
AXA PPP	2019	29	4.95%	65.41%	26.07%	3.57%
AXA PPP	2019	29	18.36%	60.02%	17.49%	14.22%
AXA PPP	2019	29	59.55%	31.84%	5.53%	3.08%
AXA PPP	2019	29	18.22%	76.26%	3.78%	1.82%
AXA PPP	2019	29	67.39%	24.02%	5.49%	3.05%
AXA PPP	2019	29	66.18%	23.94%	6.52%	3.36%
AXA PPP	2019	29	88.55%	7.76%	1.89%	1.36%
AXA PPP	2019	29	46.02%	45.37%	5.61%	3.01%
AXA PPP	2019	29	82.11%	12.03%	3.57%	2.09%
AXA PPP	2019	29	78.17%	11.48%	5.62%	3.75%
AXA PPP	2019	29	17.46%	77.07%	3.69%	1.78%
AXA PPP	2019	29	17.11%	77.02%	3.69%	1.78%
AXA PPP	2019	29	47.10%	46.34%	3.87%	2.24%
AXA UK	2019	33	16.75%	77.30%	3.59%	1.95%
AXA UK	2019	33	57.84%	31.42%	6.31%	3.29%
AXA UK	2019	33	86.79%	9.81%	1.95%	1.43%
AXA UK	2019	33	3.78%	78.51%	15.08%	2.63%
AXA UK	2019	33	7.82%	82.02%	7.94%	2.21%
AXA UK	2019	33	27.28%	62.89%	6.29%	5.53%
AXA UK	2019	33	39.18%	55.38%	3.02%	2.06%
AXA UK	2019	33	28.16%	59.73%	8.73%	3.38%
AXA UK	2019	33	72.55%	19.53%	4.80%	3.12%
AXA UK	2019	33	18.96%	76.17%	3.17%	1.70%
AXA UK	2019	33	7.48%	86.79%	4.03%	1.69%
AXA UK	2019	33	68.13%	26.63%	2.60%	2.16%
AXA UK	2019	33	7.17%	73.88%	15.85%	3.69%
BUFA Ins Ltd	2019	37	73.22%	3.59%	9.30%	13.89%
BUFA Ins Ltd	2019	37	74.12%	4.17%	15.14%	4.97%
BUFA Ins Ltd	2019	37	13.49%	4.06%	77.23%	5.22%
BUFA Ins Ltd	2019	37	30.18%	5.86%	45.98%	17.98%
BUFA Ins Ltd	2019	37	41.10%	11.11%	34.62%	12.95%
BUFA Ins Ltd	2019	37	71.60%	4.84%	10.06%	13.50%
BUFA Ins Ltd	2019	37	52.79%	5.82%	20.11%	20.88%
BUFA Ins Ltd	2019	37	62.19%	4.81%	22.75%	10.23%
BUFA Ins Ltd	2019	37	61.55%	5.51%	17.44%	12.70%
BUFA Ins Ltd	2019	37	37.12%	5.96%	35.48%	21.31%
BUFA Ins Ltd	2019	37	48.78%	6.83%	23.21%	21.18%
BUFA Ins Ltd	2019	37	76.52%	7.15%	8.69%	7.67%
BUFA Ins Ltd	2019	37	14.68%	4.52%	94.71%	71.33%
Ecological Ins Office	2019	41	2.69%	26.82%	35.15%	35.35%
Ecological Ins Office	2019	41	4.87%	26.22%	29.22%	27.27%
Ecological Ins Office	2019	41	2.12%	5.68%	73.75%	18.45%
Ecological Ins Office	2019	41	6.52%	39.94%	27.22%	26.52%
Ecological Ins Office	2019	41	4.77%	35.39%	38.45%	21.69%
Ecological Ins Office	2019	41	4.03%	57.18%	11.83%	26.99%
Ecological Ins Office	2019	41	6.26%	35.13%	21.18%	17.43%
Ecological Ins Office	2019	41	3.94%	23.48%	9.48%	63.32%
Ecological Ins Office	2019	41	4.58%	19.03%	20.67%	15.51%
Ecological Ins Office	2019	41	2.98%	68.41%	12.17%	15.96%
Ecological Ins Office	2019	41	3.91%	42.38%	34.04%	18.66%
Ecological Ins Office	2019	41	3.45%	71.43%	6.40%	18.78%
Ecological Ins Office	2019	41	2.48%	46.38%	30.46%	20.05%
Direct Line Group	2019	45	3.61%	9.20%	16.34%	70.84%
Direct Line Group	2019	45	5.41%	29.11%	22.86%	43.52%
Direct Line Group	2019	45	2.70%	5.34%	9.04%	83.02%
Direct Line Group	2019	45	4.22%	12.21%	13.17%	70.49%
Direct Line Group	2019	45	6.53%	7.05%	19.70%	66.77%
Direct Line Group	2019	45	3.31%	4.65%	8.07%	84.00%
Direct Line Group	2019	45	3.52%	3.62%	12.59%	80.28%
Direct Line Group	2019	45	5.40%	9.62%	43.50%	41.50%
Direct Line Group	2019	45	5.69%	35.47%	33.30%	25.54%
Direct Line Group	2019	45	5.17%	12.86%	26.82%	55.15%
Direct Line Group	2019	45	3.64%	15.32%	30.79%	50.20%
Direct Line Group	2019	45	6.97%	25.08%	17.50%	50.44%
Direct Line Group	2019	45	5.02%	6.79%	56.71%	71.56%
esure Group	2019	49	20.12%	14.56%	12.89%	32.44%
esure Group	2019	49	5.16%	7.51%	13.21%	73.51%
esure Group	2019	49	4.36%	4.92%	17.04%	73.67%
esure Group	2019	49	13.49%	6.27%	46.04%	33.75%
esure Group	2019	49	11.28%	7.61%	46.41%	34.69%
esure Group	2019	49	0.92%	2.16%	2.69%	94.23%
esure Group	2019	49	3.97%	9.75%	9.79%	76.58%
esure Group	2019	49	1.58%	6.60%	12.11%	79.79%
esure Group	2019	49	6.91%	7.12%	13.18%	72.89%
esure Group	2019	49	20.37%	13.85%	22.54%	43.54%
esure Group	2019	49	11.48%	11.72%	43.75%	33.65%
esure Group	2019	49	4.92%	1.00%	30.38%	56.72%
esure Group	2019	49	2.68%	7.42%	4.29%	86.36%
Fidus Underwriting Ltd	2019	53	0.86%	0.75%	97.82%	0.77%
Fidus Underwriting Ltd	2019	53	0.46%	0.92%	97.91%	0.74%
Fidus Underwriting Ltd	2019	53	0.69%	1.00%	97.28%	1.01%
Fidus Underwriting Ltd	2019	53	0.65%	1.74%	95.75%	1.56%
Fidus Underwriting Ltd	2019	53	1.43%	2.51%	93.67%	3.39%
Fidus Underwriting Ltd	2019	53	0.70%	0.81%	97.46%	1.03%
Fidus Underwriting Ltd	2019	53	0.61%	0.88%	97.67%	0.79%
Fidus Underwriting Ltd	2019	53	0.61%	0.71%	97.40%	0.80%
Fidus Underwriting Ltd	2019	53	1.58%	3.62%	90.69%	4.11%
Fidus Underwriting Ltd	2019	53	0.83%	1.05%	96.71%	1.43%
Fidus Underwriting Ltd	2019	53	0.74%	1.00%	96.97%	1.28%
Fidus Underwriting Ltd	2019	53	0.79%	1.24%	96.49%	1.48%
Fidus Underwriting Ltd	2019	53	3.94%	4.89%	83.23%	8.03%

7. Code and data

GitHub Repositories: <https://github.com/simoneSantoni/risk-in-insurance-industry>